

Locality from Zeros

1 Seven-point residue on X_{16}

We now illustrate the residue strategy at seven points. Assume again that no higher-order poles are present.

Take the parent residue

$$X_{16} = 0. \quad (1)$$

The only chords crossing X_{16} are

$$X_{27}, \quad X_{37}, \quad X_{47}, \quad X_{57}. \quad (2)$$

Thus every non-locality relative to the parent pole involves label 7.

A seven-point cubic term has four propagators. After taking the residue on X_{16} , three propagators remain. We organize the residue by pulling out explicitly all poles that cross X_{16} :

$$\text{Res}_{X_{16}=0} B_7 = B_6^{(3)}(1, 2, 3, 4, 5, 6) + \sum_{i=2}^5 \frac{C_i^{(2)}(1, 2, 3, 4, 5, 6)}{X_{i7}} + \sum_{2 \leq i < j \leq 5} \frac{C_{ij}^{(1)}(1, 2, 3, 4, 5, 6)}{X_{i7} X_{j7}} + \sum_{2 \leq i < j < k \leq 5} \frac{C_{ijk}^{(0)}(1, 2, 3, 4, 5, 6)}{X_{i7} X_{j7} X_{k7}}. \quad (3)$$

The notation is chosen so that $C^{(k)}$ has k remaining denominator factors. Thus:

$$C_i^{(2)} \quad \text{is a six-point two-pole function,} \quad (4)$$

$$C_{ij}^{(1)} \quad \text{is a six-point one-pole function,} \quad (5)$$

and

$$C_{ijk}^{(0)} \quad \text{is a constant.} \quad (6)$$

Crucially, all these functions depend only on the lower labels

$$1, 2, 3, 4, 5, 6. \quad (7)$$

This is again useful because it can enable inductive reasoning. The explicit factors X_{i7} encode the non-locality relative to the parent pole X_{16} .

Therefore, to prove locality near this parent pole, we need to show

$$C_i^{(2)} = 0, \quad C_{ij}^{(1)} = 0, \quad C_{ijk}^{(0)} = 0. \quad (8)$$

This removes every term containing a pole crossing X_{16} .

To prove full locality, this procedure must be repeated for every possible parent pole.

The question is then how to prove that all C functions are eliminated by zero conditions.

2 The induction step at seven points

We write the planar variables as $X_{i,j}$, with cyclic labels understood. Let Z_a denote the ordinary zero associated with the vanishing of the a -th row of the mesh. We shall use the following notion of strictness.

Definition 1. *Let M be a set of poles. A zero Z_a is called strict with respect to M if the system*

$$X = 0 \quad (X \in M), \quad Z_a = 0$$

does not force any planar pole outside M to vanish.

The induction hypothesis at n points is the following.

1. The top-degree object $B_n^{(n-3)}$ is fixed projectively by $n - 1$ zeros:

$$B_n^{(n-3)} = \lambda A_n.$$

2. The boundary nonlocal object $C_n^{(n-4)}$ is killed by $n - 2$ zeros:

$$C_n^{(n-4)} = 0.$$

3. The lower nonlocal objects are killed by $k + 1$ zeros:

$$C_n^{(k)} = 0, \quad k < n - 4.$$

We check the induction step at seven points, assuming the corresponding six-point statements.

2.1 Induction closes on the lower C -functions

We first check that the proposed counting closes on the C -functions at seven points.

We organize the proof by how a parent pole P interacts with the available zeros. After taking a parent-pole residue, one expands in the poles crossing P . The coefficients in this crossing expansion are the objects to which the lower-point induction hypothesis applies.

There are three types of parent poles.

1. P does not overlap with the available zeros. Then all available zeros survive strictly on the coefficients in the crossing expansion.
2. P overlaps with one available zero. Then each coefficient loses at most one zero, which is exactly what is allowed by the residue induction.
3. P effectively removes two available zeros. Then some coefficient in the crossing expansion can have one fewer strict zero than needed. This is the only delicate case. It must be handled by a saturated-active zero, activated by a residue which has already been eliminated.

We illustrate this mechanism at six points. The six-point planar poles are

$$\begin{aligned} &X_{1,3}, X_{1,4}, X_{1,5}, \\ &X_{2,4}, X_{2,5}, X_{2,6}, \\ &X_{3,5}, X_{3,6}, X_{4,6}. \end{aligned}$$

2.2 Parent-pole classification and the six-point test

We now explain the induction mechanism at six points, assuming the corresponding five-point statements. We check what happens after taking a parent-pole residue and expanding in the poles crossing the parent pole.

At five points, the relevant induction statements are:

$$C_5^{(0)} = 0 \quad \text{from one zero,}$$

and

$$C_5^{(1)} = 0 \quad \text{from three zeros.}$$

The second statement is the five-point boundary case, since $1 = 5 - 4$.

We use the following threefold classification. A parent pole P is:

1. *good* if all coefficients in the crossing expansion inherit strictly more zeros than needed;
2. *marginal* if all coefficients inherit exactly the number of zeros needed;
3. *bad* if the top coefficient in the crossing expansion inherits one zero fewer than needed. Such a pole must be treated by a saturated-active zero coming from a full residue that has already been eliminated.

The six-point planar poles are

$$\begin{aligned} &X_{1,3}, X_{1,4}, X_{1,5}, \\ &X_{2,4}, X_{2,5}, X_{2,6}, \\ &X_{3,5}, X_{3,6}, X_{4,6}. \end{aligned}$$

2.2.1 $C_6^{(1)}$

Let

$$F = C_6^{(1)}, \quad F|_{Z_1} = F|_{Z_2} = 0.$$

Since $1 < 6 - 4$, the refined induction statement predicts that two zeros kill F .

A residue of F is a constant. Hence a parent pole is eliminated as soon as its residue inherits one strict zero. The strict-zero table is

parent pole P	strict zeros among Z_1, Z_2	type
$X_{1,3}$	$\emptyset$	bad
$X_{1,4}$	Z_1, Z_2	good
$X_{1,5}$	Z_1, Z_2	good
$X_{2,4}$	Z_1	marginal
$X_{2,5}$	Z_1, Z_2	good
$X_{2,6}$	Z_2	marginal
$X_{3,5}$	Z_1, Z_2	good
$X_{3,6}$	Z_1, Z_2	good
$X_{4,6}$	Z_1, Z_2	good

Thus all good and marginal parent poles are eliminated immediately:

$$\text{Res}_{P=0} F = 0 \quad \text{for } P \neq X_{1,3}.$$

The only bad parent pole is $X_{1,3}$. Its residue is

$$\text{Res}_{X_{1,3}=0} F = C_{1,3}^{(0)}.$$

Neither Z_1 nor Z_2 is strict for this residue. For example, using

$$Z_1 : \begin{cases} X_{1,3} + X_{2,4} - X_{1,4} = 0, \\ X_{1,4} + X_{2,5} - X_{1,5} - X_{2,4} = 0, \\ X_{1,5} + X_{2,6} - X_{2,5} = 0, \end{cases}$$

and imposing $X_{1,3} = 0$, one obtains

$$X_{1,4} = X_{2,4}, \quad X_{2,5} = X_{1,5}, \quad X_{2,6} = 0.$$

Thus Z_1 is polluted by $X_{2,6}$. But the full residue

$$\text{Res}_{X_{2,6}=0} F$$

has already vanished, because $X_{2,6}$ is marginal and has Z_2 strict. Hence Z_1 is saturated-active for the $X_{1,3}$ -residue. Therefore

$$\text{Res}_{X_{1,3}=0} F = 0.$$

All residues of F vanish, and therefore

$$C_6^{(1)} = 0.$$

Equivalently, Z_2 may be used instead: imposing $X_{1,3} = 0$ and $Z_2 = 0$ forces $X_{2,4} = 0$, and the full residue at $X_{2,4} = 0$ has already vanished.

2.2.2 $C_6^{(2)}$

Let

$$F = C_6^{(2)}, \quad F|_{Z_1} = F|_{Z_2} = F|_{Z_3} = F|_{Z_4} = 0.$$

Since $2 = 6 - 4$, this is the six-point boundary case. The refined induction statement predicts that four zeros kill F .

A parent-pole residue has degree one. It is still a six-point expression, so we expand it in the poles crossing the parent pole. The coefficients in this crossing expansion are five-point objects. The constants $C^{(0)}$ are killed by one zero, while the top coefficient $G^{(1)}$ is a five-point boundary object and requires three zeros.

The strict-zero table for the top coefficient of each parent-pole residue is

parent pole P	strict zeros of $G_P^{(1)}$	type
$X_{1,3}$	Z_3, Z_4	bad
$X_{1,4}$	Z_1, Z_2, Z_3, Z_4	good
$X_{1,5}$	Z_1, Z_2, Z_3, Z_4	good
$X_{2,4}$	Z_1, Z_4	bad
$X_{2,5}$	Z_1, Z_2, Z_3, Z_4	good
$X_{2,6}$	Z_2, Z_3, Z_4	marginal
$X_{3,5}$	Z_1, Z_2	bad
$X_{3,6}$	Z_1, Z_2, Z_3, Z_4	good
$X_{4,6}$	Z_1, Z_2, Z_3	marginal

The good and marginal poles will be eliminated first. The bad poles form the chain

$$X_{4,6} \longrightarrow X_{3,5} \longrightarrow X_{2,4} \longrightarrow X_{1,3}.$$

Here $X_{4,6}$ is not bad; it is marginal and provides the first full residue needed to activate the chain.

A good parent pole: $X_{2,5}$. Consider

$$P = X_{2,5}.$$

The poles crossing $X_{2,5}$ are

$$X_{1,3}, \quad X_{1,4}, \quad X_{3,6}, \quad X_{4,6}.$$

Thus

$$\begin{aligned} \text{Res}_{X_{2,5}=0} F &= G_{2,5}^{(1)} + \frac{C_{2,5;1,3}^{(0)}}{X_{1,3}} + \frac{C_{2,5;1,4}^{(0)}}{X_{1,4}} \\ &\quad + \frac{C_{2,5;3,6}^{(0)}}{X_{3,6}} + \frac{C_{2,5;4,6}^{(0)}}{X_{4,6}}. \end{aligned}$$

The constants are eliminated first. Their strict zeros are

coefficient	strict zeros among Z_1, Z_2, Z_3, Z_4
$C_{2,5;1,3}^{(0)}$	Z_3, Z_4
$C_{2,5;1,4}^{(0)}$	Z_1, Z_2, Z_3, Z_4
$C_{2,5;3,6}^{(0)}$	Z_1, Z_2, Z_3, Z_4
$C_{2,5;4,6}^{(0)}$	Z_1, Z_2, Z_3

Each constant has at least one strict zero, so all constants vanish by the five-point $C_5^{(0)}$ statement. We are left with

$$\text{Res}_{X_{2,5}=0} F = G_{2,5}^{(1)}.$$

The top coefficient has

$$G_{2,5}^{(1)} : \quad Z_1, Z_2, Z_3, Z_4$$

strict. This is more than the three zeros required for a five-point boundary $C_5^{(1)}$. Hence

$$G_{2,5}^{(1)} = 0,$$

and therefore

$$\text{Res}_{X_{2,5}=0} F = 0.$$

A marginal parent pole: $X_{4,6}$. Now consider

$$P = X_{4,6}.$$

The poles crossing $X_{4,6}$ are

$$X_{1,5}, \quad X_{2,5}, \quad X_{3,5}.$$

Thus

$$\text{Res}_{X_{4,6}=0} F = G_{4,6}^{(1)} + \frac{C_{4,6;1,5}^{(0)}}{X_{1,5}} + \frac{C_{4,6;2,5}^{(0)}}{X_{2,5}} + \frac{C_{4,6;3,5}^{(0)}}{X_{3,5}}.$$

Again the constants vanish first. Their strict zeros are

coefficient	strict zeros among Z_1, Z_2, Z_3, Z_4
$C_{4,6;1,5}^{(0)}$	Z_1, Z_2, Z_3
$C_{4,6;2,5}^{(0)}$	Z_1, Z_2, Z_3
$C_{4,6;3,5}^{(0)}$	Z_1, Z_2, Z_4

so all constants vanish. The top coefficient has exactly three strict zeros:

$$G_{4,6}^{(1)} : \quad Z_1, Z_2, Z_3.$$

This is exactly the number required for the five-point boundary statement. Therefore

$$G_{4,6}^{(1)} = 0, \quad \text{Res}_{X_{4,6}=0} F = 0.$$

This marginal residue will now activate the next bad parent pole.

The first bad parent pole: $X_{3,5}$. Consider

$$P = X_{3,5}.$$

The poles crossing $X_{3,5}$ are

$$X_{1,4}, \quad X_{2,4}, \quad X_{4,6}.$$

Thus

$$\text{Res}_{X_{3,5}=0} F = G_{3,5}^{(1)} + \frac{C_{3,5;1,4}^{(0)}}{X_{1,4}} + \frac{C_{3,5;2,4}^{(0)}}{X_{2,4}} + \frac{C_{3,5;4,6}^{(0)}}{X_{4,6}}.$$

The constants are killed first:

coefficient	strict zeros among Z_1, Z_2, Z_3, Z_4
$C_{3,5;1,4}^{(0)}$	Z_1, Z_2
$C_{3,5;2,4}^{(0)}$	Z_1, Z_3
$C_{3,5;4,6}^{(0)}$	Z_1, Z_2, Z_4

Thus all constants vanish, and

$$\text{Res}_{X_{3,5}=0} F = G_{3,5}^{(1)}.$$

The top coefficient has only

$$G_{3,5}^{(1)} : \quad Z_1, Z_2$$

strict, one fewer than the three zeros required.

The missing zero is Z_4 , which is not strict but is now active. Indeed,

$$Z_4 : \begin{cases} X_{4,6} + X_{1,5} - X_{1,4} = 0, \\ X_{1,4} + X_{2,5} - X_{1,5} - X_{2,4} = 0, \\ X_{2,4} + X_{3,5} - X_{2,5} = 0. \end{cases}$$

Together with $X_{3,5} = 0$, these equations force

$$X_{4,6} = 0.$$

But the full residue $\text{Res}_{X_{4,6}=0} F$ has already vanished. Hence Z_4 is saturated-active for $G_{3,5}^{(1)}$. Therefore

$$G_{3,5}^{(1)}$$

has the three usable zeros

$$Z_1, \quad Z_2, \quad Z_4 \text{ active.}$$

Thus

$$\text{Res}_{X_{3,5}=0} F = 0.$$

The second bad parent pole: $X_{2,4}$. The poles crossing $X_{2,4}$ are

$$X_{1,3}, \quad X_{3,5}, \quad X_{3,6}.$$

So

$$\text{Res}_{X_{2,4}=0} F = G_{2,4}^{(1)} + \frac{C_{2,4;1,3}^{(0)}}{X_{1,3}} + \frac{C_{2,4;3,5}^{(0)}}{X_{3,5}} + \frac{C_{2,4;3,6}^{(0)}}{X_{3,6}}.$$

The constants have strict zeros

coefficient	strict zeros among Z_1, Z_2, Z_3, Z_4
$C_{2,4;1,3}^{(0)}$	Z_2, Z_4
$C_{2,4;3,5}^{(0)}$	Z_1, Z_3
$C_{2,4;3,6}^{(0)}$	Z_1, Z_4

so they vanish. Hence

$$\text{Res}_{X_{2,4}=0} F = G_{2,4}^{(1)}.$$

The top coefficient has only

$$G_{2,4}^{(1)} : \quad Z_1, Z_4$$

strict. The zero Z_3 is active, because $X_{2,4} = 0$ together with $Z_3 = 0$ forces

$$X_{3,5} = 0.$$

Since the full residue at $X_{3,5} = 0$ has just been eliminated, Z_3 is saturated-active. Thus

$$\text{Res}_{X_{2,4}=0} F = 0.$$

The final bad parent pole: $X_{1,3}$. The poles crossing $X_{1,3}$ are

$$X_{2,4}, \quad X_{2,5}, \quad X_{2,6}.$$

Thus

$$\text{Res}_{X_{1,3}=0} F = G_{1,3}^{(1)} + \frac{C_{1,3;2,4}^{(0)}}{X_{2,4}} + \frac{C_{1,3;2,5}^{(0)}}{X_{2,5}} + \frac{C_{1,3;2,6}^{(0)}}{X_{2,6}}.$$

The constants vanish:

coefficient	strict zeros among Z_1, Z_2, Z_3, Z_4
$C_{1,3;2,4}^{(0)}$	Z_2, Z_4
$C_{1,3;2,5}^{(0)}$	Z_3, Z_4
$C_{1,3;2,6}^{(0)}$	Z_1, Z_3, Z_4

Therefore

$$\text{Res}_{X_{1,3}=0} F = G_{1,3}^{(1)}.$$

The top coefficient has only

$$G_{1,3}^{(1)} : \quad Z_3, Z_4$$

strict. The zero Z_2 is active, because $X_{1,3} = 0$ together with $Z_2 = 0$ forces

$$X_{2,4} = 0.$$

Since the full residue at $X_{2,4} = 0$ has already vanished, Z_2 is saturated-active. Hence

$$\text{Res}_{X_{1,3}=0} F = 0.$$

Thus all parent-pole residues of $F = C_6^{(2)}$ vanish, and therefore

$$C_6^{(2)} = 0.$$

The structure is now clear. Good poles are eliminated first by surplus strict zeros. Marginal poles are eliminated next by exactly the required number of strict zeros. These marginal residues then supply the saturated-active zeros needed to eliminate the bad chain

$$X_{4,6} \longrightarrow X_{3,5} \longrightarrow X_{2,4} \longrightarrow X_{1,3}.$$

3 Propagating the zero residue: from $X_{1,5}$ to $X_{4,6}$

The previous section established that all C functions vanish. This means the residue of $X_{1,5}$ can be uniquely fixed by induction to be $R_{1,5} = \lambda A_5(1, 2, 3, 4, 5)$.

We now pass from the vanishing of the first parent-pole residue to the vanishing of the next residue, on $X_{4,6}$, which is itself a "bad parent pole", because it overlaps two zeros, 4 and 5. The previous pole was a marginal parent pole. Let

$$D_6 := B_6^{(3)} - A_6.$$

Assume that the normalization has been chosen so that

$$\text{Res}_{X_{1,5}=0} D_6 = 0.$$

We shall prove this implies

$$\text{Res}_{X_{4,6}=0} D_6 = 0.$$

The zero which relates these two poles is Z_5 . At six points,

$$Z_5 : \begin{cases} X_{1,5} + X_{2,6} - X_{2,5} = 0, \\ X_{2,5} + X_{3,6} - X_{2,6} - X_{3,5} = 0, \\ X_{3,5} + X_{4,6} - X_{3,6} = 0. \end{cases}$$

Solving these equations gives

$$\begin{aligned} X_{2,6} &= X_{2,5} - X_{1,5}, \\ X_{3,6} &= X_{3,5} - X_{1,5}, \end{aligned}$$

and therefore

$$X_{4,6} = -X_{1,5}.$$

Thus Z_5 identifies the $X_{4,6}$ -residue with the already killed $X_{1,5}$ -residue. However, before using this relation, we must remove the remaining label-5 poles from the $X_{4,6}$ -residue.

Set

$$R_{4,6} := \operatorname{Res}_{X_{4,6}=0} D_6.$$

The poles crossing $X_{4,6}$ are

$$X_{1,5}, \quad X_{2,5}, \quad X_{3,5}.$$

The $X_{1,5}$ -sector has already been removed, because

$$\operatorname{Res}_{X_{1,5}=0} D_6 = 0.$$

Hence the remaining crossing expansion is

$$R_{4,6} = G_{4,6}^{(2)} + \frac{C_{4,6;2,5}^{(1)}}{X_{2,5}} + \frac{C_{4,6;3,5}^{(1)}}{X_{3,5}} + \frac{C_{4,6;2,5;3,5}^{(0)}}{X_{2,5}X_{3,5}}.$$

We eliminate this expansion triangularly:

$$C^{(0)} \longrightarrow C^{(1)} \longrightarrow G^{(2)}.$$

First consider the constant coefficient

$$C_{4,6;2,5;3,5}^{(0)}.$$

Its pole set is

$$M = \{X_{4,6}, X_{2,5}, X_{3,5}\}.$$

The zero Z_1 is strict for this pole set. Indeed,

$$Z_1 : \begin{cases} X_{1,3} + X_{2,4} - X_{1,4} = 0, \\ X_{1,4} + X_{2,5} - X_{1,5} - X_{2,4} = 0, \\ X_{1,5} + X_{2,6} - X_{2,5} = 0, \end{cases}$$

and after imposing

$$X_{4,6} = X_{2,5} = X_{3,5} = 0,$$

these equations impose only linear relations among

$$X_{1,3}, X_{2,4}, X_{1,4}, X_{1,5}, X_{2,6},$$

and force no additional planar pole outside M to vanish. Hence

$$C_{4,6;2,5;3,5}^{(0)} = 0.$$

Next consider the one-pole coefficient

$$C_{4,6;2,5}^{(1)}.$$

Its pole set is

$$M = \{X_{4,6}, X_{2,5}\}.$$

The strict zeros are

$$Z_1, \quad Z_2, \quad Z_3.$$

Let us record the check. With $X_{4,6} = X_{2,5} = 0$:

$$Z_1 : \begin{cases} X_{1,3} + X_{2,4} - X_{1,4} = 0, \\ X_{1,4} - X_{1,5} - X_{2,4} = 0, \\ X_{1,5} + X_{2,6} = 0, \end{cases}$$

which forces no pole to vanish.

$$Z_2 : \begin{cases} X_{2,4} + X_{3,5} = 0, \\ X_{3,6} - X_{2,6} - X_{3,5} = 0, \\ X_{2,6} + X_{1,3} - X_{3,6} = 0, \end{cases}$$

which gives only linear relations, for example

$$X_{2,4} + X_{1,3} = 0,$$

and forces no individual planar pole to vanish.

Finally,

$$Z_3 : \begin{cases} X_{3,5} + X_{4,6} - X_{3,6} = 0, \\ X_{3,6} + X_{1,4} - X_{1,3} - X_{4,6} = 0, \\ X_{1,3} + X_{2,4} - X_{1,4} = 0, \end{cases}$$

and with $X_{4,6} = 0$ this gives

$$X_{3,6} = X_{3,5}, \quad X_{1,4} = X_{1,3} - X_{3,6}, \quad X_{1,4} = X_{1,3} + X_{2,4},$$

again forcing no pole to vanish. Therefore

$$C_{4,6;2,5}^{(1)}$$

has three strict zeros. It is a five-point boundary object of type $C_5^{(1)}$, so the five-point hypothesis gives

$$C_{4,6;2,5}^{(1)} = 0.$$

Similarly, consider

$$C_{4,6;3,5}^{(1)}.$$

Its pole set is

$$M = \{X_{4,6}, X_{3,5}\}.$$

The strict zeros are

$$Z_1, \quad Z_2, \quad Z_4.$$

Indeed, Z_1 does not involve $X_{3,5}$ or $X_{4,6}$ in a way that forces any new pole to vanish. For Z_2 , imposing $X_{3,5} = 0$ gives

$$\begin{cases} X_{2,4} - X_{2,5} = 0, \\ X_{2,5} + X_{3,6} - X_{2,6} = 0, \\ X_{2,6} + X_{1,3} - X_{3,6} = 0, \end{cases}$$

which only implies the linear relation $X_{2,4} + X_{1,3} = 0$. For Z_4 ,

$$Z_4 : \begin{cases} X_{4,6} + X_{1,5} - X_{1,4} = 0, \\ X_{1,4} + X_{2,5} - X_{1,5} - X_{2,4} = 0, \\ X_{2,4} + X_{3,5} - X_{2,5} = 0, \end{cases}$$

and imposing $X_{4,6} = X_{3,5} = 0$ gives

$$X_{1,4} = X_{1,5}, \quad X_{2,5} = X_{2,4},$$

with no new pole forced to vanish. Hence

$$C_{4,6;3,5}^{(1)}$$

has three strict zeros, and therefore

$$C_{4,6;3,5}^{(1)} = 0$$

by the five-point boundary statement.

Thus all non-top pieces in the $X_{4,6}$ -residue have vanished, and we are left with

$$R_{4,6} = G_{4,6}^{(2)}.$$

This is the label-deleted top piece: it contains no remaining label-5 poles.

Now we use Z_5 . Near the two relevant poles,

$$D_6 = \frac{R_{1,5}}{X_{1,5}} + \frac{R_{4,6}}{X_{4,6}} + \cdots,$$

where

$$R_{1,5} := \operatorname{Res}_{X_{1,5}=0} D_6 = 0.$$

On Z_5 , we have

$$X_{4,6} = -X_{1,5}.$$

Since D_6 vanishes on Z_5 , the residues must cancel on this locus:

$$R_{4,6}|_{Z_5} = R_{1,5}|_{Z_5}.$$

But $R_{1,5} = 0$, hence

$$R_{4,6}|_{Z_5} = 0.$$

After the label-5 poles $X_{2,5}$ and $X_{3,5}$ have been removed, the remaining object

$$R_{4,6} = G_{4,6}^{(2)}$$

is independent of label-5 kinematics. Therefore imposing Z_5 only solves for the deleted label-5 variables and imposes no further condition on $G_{4,6}^{(2)}$. Hence

$$R_{4,6}|_{Z_5} = R_{4,6}.$$

We conclude

$$\text{Res}_{X_{4,6}=0} D_6 = 0.$$

Thus the zero residue has propagated:

$$\text{Res}_{X_{1,5}=0} D_6 = 0 \implies \text{Res}_{X_{4,6}=0} D_6 = 0.$$

The argument can be continued for all remaining poles of D_6 , finally showing $D_6 = 0$, hence $B_6 = A_6$.